



Self-Healing Mortar with Superabsorbent Polymers: A Fracture Mechanics Study of Dry and Calcium Loaded Systems

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1. Motivation

- Cracking reduces durability and accelerates maintenance needs
- Controlling crack growth can extend service life
- Reduced repairs lower material consumption and construction waste
- Fewer interventions can reduce lifecycle CO₂ emissions
- Self-healing additives support more sustainable concrete systems



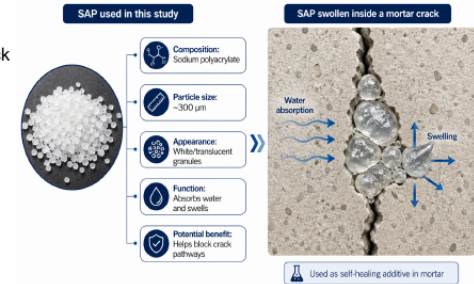
2. Objective

- Study the effect of dry SAP and limewater-loaded SAP on mortar fracture resistance using SCB testing
- Evaluate crack reopening after pre-cracking at 60% of peak load and healing from Day 7 to Day 14
- Compare retained strength, energy absorption, and fracture-related response among Control, Dry SAP, and Lime SAP mixtures
- Determine whether calcium-loaded SAP improves self-healing performance beyond dry SAP
- Assess SAP-modified mortar using fracture-based behavior

3. Proposed Solution – Super Absorbent Polymer (SAP)

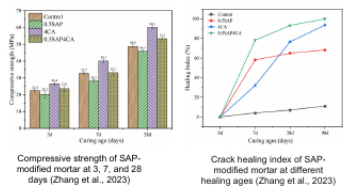
- SAP is introduced as an internal self-healing additive in mortar
- SAP absorbs water and swells within the matrix
- During cracking, SAP can re-swell in the crack region
- Swollen SAP may reduce crack connectivity and moisture ingress
- Released moisture supports hydration and crack sealing
- In limewater-loaded SAP systems, released Ca²⁺ may form CaCO₃ within cracks
- Dry SAP and Lime SAP were compared with the Control mix

SAP used in this study and why it matters

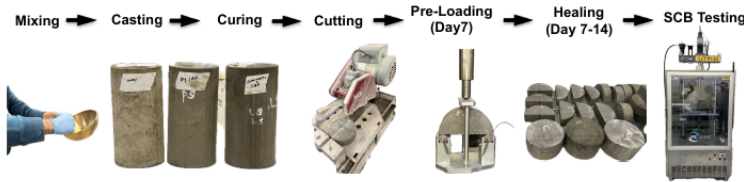


4. Literature Review

- SAP improves internal curing and supports self-healing in cementitious materials (Cheng et al., 2022).
- Released moisture from SAP can promote continued hydration and additional C-S-H formation (Cheng et al., 2022).
- SAP may influence compressive strength depending on dosage, size, and curing condition (Niu et al., 2024).



5. Methodology



Specimen Preparation

- Three mortar mixes were prepared: Control, Dry SAP, and Lime SAP
- Mortar was cast into cylindrical specimens and cured
- Cylinders were cut into semi-circular bending (SCB) specimens for testing

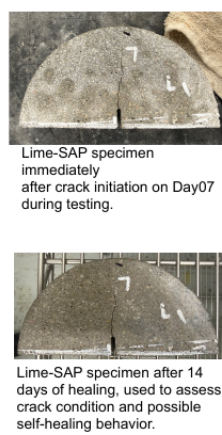
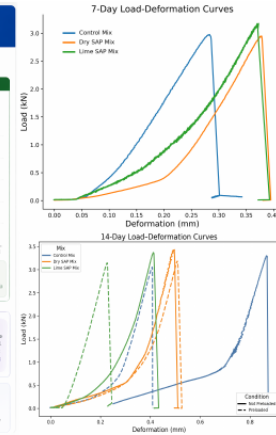
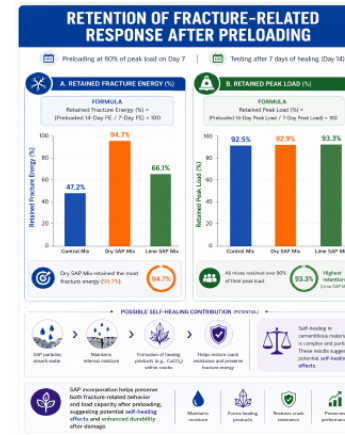
Testing Procedure

- Pre-loading (Day 7): Specimens were loaded to 60% of peak load to induce controlled crack damage
- Healing (Day 7–14): Damaged specimens were exposed to moisture to promote healing
- SCB testing: Semi-circular bending tests were performed to evaluate fracture behavior and post-damage response

Key Parameters

- Test Method: Semi-Circular Bending (SCB)
- Pre-loading Level: 60% of peak load at Day 7
- Evaluated Response: Peak load, deformation, fracture energy, ductility, and retention

6. Results



7. Conclusions

- ✓ Preloaded mortar specimens treated with SAP showed a slight in retained peak load and fracture energy ratio compared with the control. Results depend on treatment conditions.
- ✓ Based on the literature, more pronounced effects are expected after 28 days.
- ✓ The research group is evaluating longer curing periods to confirm SAP effectiveness. Additionally, the dosage may need to be revised.

8. Key Takeaways

- Fracture Resistance Potential:** SAP-modified mortars showed potential for increase fracture resistance.
- Expected Performance Growth:** The fracture response is expected to grow significantly with a higher curing period.
- Promising Self-Healing Additive:** SAP shows promising self-healing potential for concrete.

9. References

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